

ETL Workflow

NorthStar Manufacturing Group

FP&A Quarterly Business Performance Review

1. Purpose

This document summarizes the Extract, Transform, Load (ETL) workflow used to prepare financial data for the FP&A Quarterly Business Performance Review. It outlines the end-to-end process from raw source files through SQL Server data preparation, validation, dimensional warehouse loading, and Power BI reporting.

2. Source Data

- General Ledger
- Budget Forecast
- Accounts Payable
- Accounts Receivable

3. Extraction

Raw Microsoft Excel source files were imported into SQL Server source tables. These tables served as the initial landing area for profiling, cleaning, and validation before any reporting transformations were applied.

4. Data Profiling

Each source dataset was profiled to assess data quality and identify issues prior to transformation. Profiling activities included:

- Row counts
- Column structure verification
- Missing value analysis
- Duplicate key detection
- Transaction and reporting date range validation
- Status and date consistency checks
- Department, cost center, and currency distribution analysis

5. Data Cleaning

Data cleaning standardized source data for analytical use. This included correcting data types, addressing invalid or inconsistent values where appropriate, applying business rules, and validating cleaned outputs before warehouse loading.

6. Warehouse Loading

Cleaned datasets were transformed into a dimensional data warehouse consisting of:

- 9 Dimension Tables
- 4 Fact Tables

The warehouse design incorporated surrogate keys, foreign key relationships, unique constraints, and indexes to support data integrity, performance, and analytical reporting.

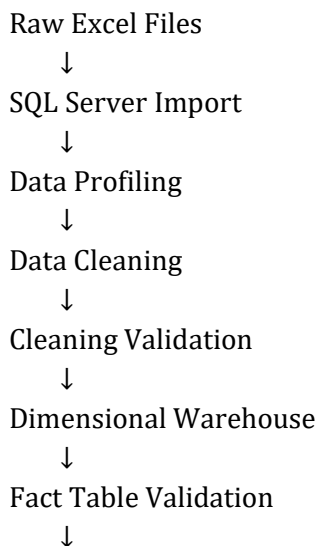
7. Fact Validation

Following warehouse loading, each fact table was validated to confirm successful population and consistency. Validation activities included row count verification, key integrity checks, and reconciliation against the transformed source data.

8. Power BI Integration

Power BI connected directly to the SQL Server data warehouse. The semantic model established one-to-many relationships between fact and dimension tables, while DAX measures were created to support financial KPIs, variance analysis, forecasting, working capital reporting, and executive dashboards.

9. Workflow Diagram



Constraints and Indexes



Power BI Semantic Model



DAX Measures



Executive Dashboard

10. Controls and Quality Checks

The ETL process incorporated multiple quality controls to improve data reliability before reporting:

- Duplicate detection
- Missing-value review
- Foreign-key verification
- Data-type correction
- Relationship validation
- Reconciliation before reporting